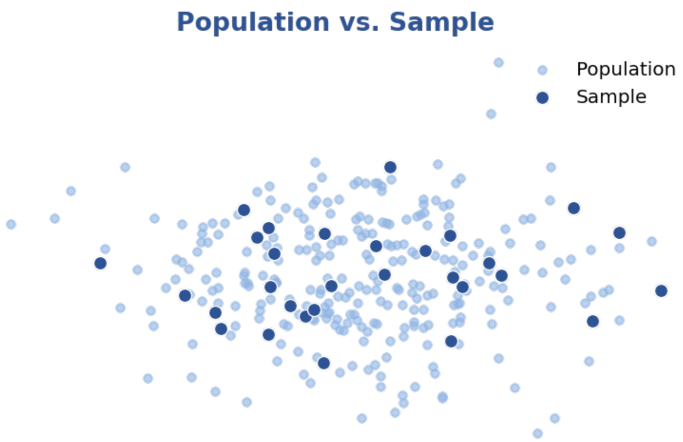
# Derivable Judgement: A Statistical Decision-Making Model

*Part A — Theoretical Foundation*

## 1. What is inferential statistics?

Inferential Statistical is the branch of statistical the uses sample data to make conclusions or prediction about the entire population.

It help research estimate population parameters and test hypotheses without collecting data from every individual



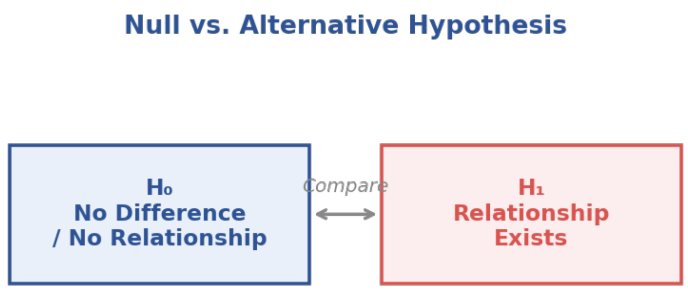

## 2. What is hypothesis testing and its components?

Hypothesis Testing is a statistical method to determine whether there is enough evidence to reject a claim about a population

It compares sample data with assumptions about the population

***Type of hypothesis testing***

1. Null Hypothesis: No relationship or no difference exits
2. Alternative Hypothesis: Relationship exits



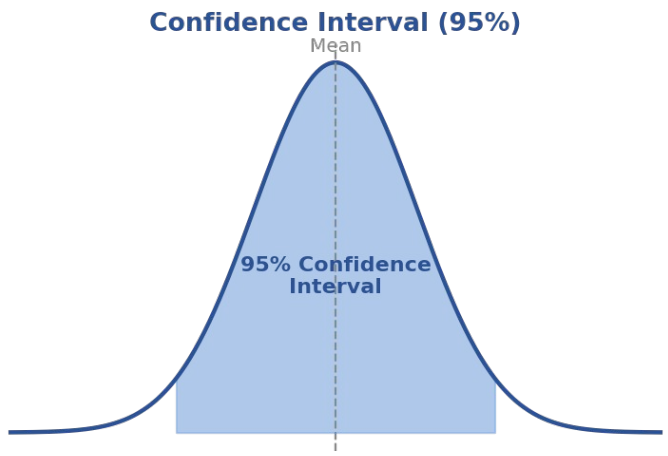

## 3. Explain confidence interval and critical value.

### *1. Confidence Interval*

Confidence interval is a range of values that likely to contain the true population parameter.

95% Confidence interval means we are 95% confident that true mean live within this range

| |
|---|
| Formula => mean += margin of error |




### *2. Critical Value*

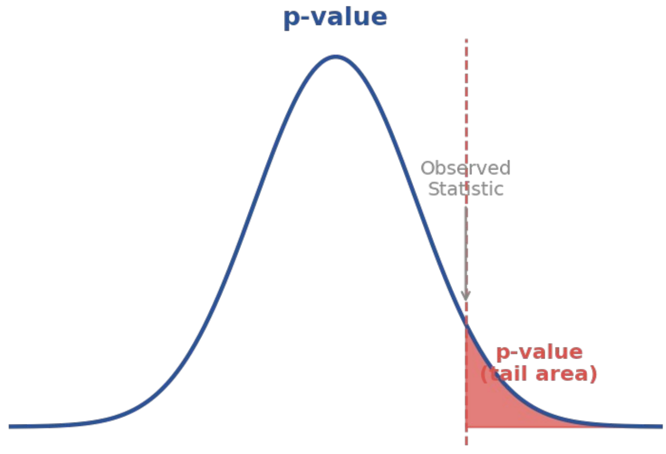
Critical value is the cutoff value to decide whether to reject the null hypothesis

## 4. Define p-value.

P-value is the probability of obtaining the observed result assuming the null hypothesis is true

***Decision:***

$p < 0.05$ => Reject H0        $p > 0.05$ => Fail to Reject H0




## 5. Differentiate Type I and Type II errors.

***1. Type I Error***

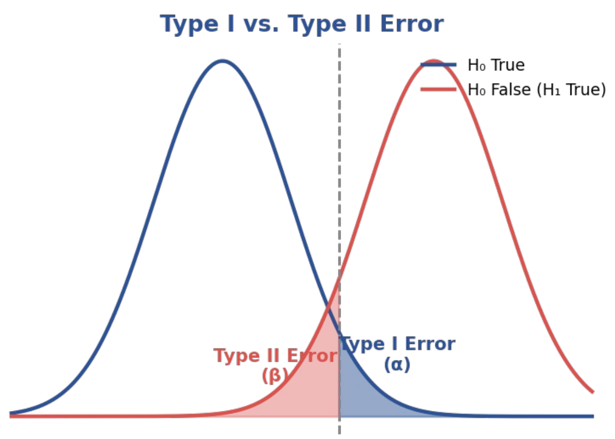
Reject True H0

False Positive

***2. Type II Error***

Accept False H0

False Negative



## 6. Brief descriptions of z-test, t-test, chi-square test, and ANOVA test.

### *1. Z-test*

Compare sample mean with population mean

Used for Large sample (n >= 30) and population standard deviation is known

### *2. T-Test*

Compare means of two groups

Used for Small sample (n < 30) and population standard deviation is unknown

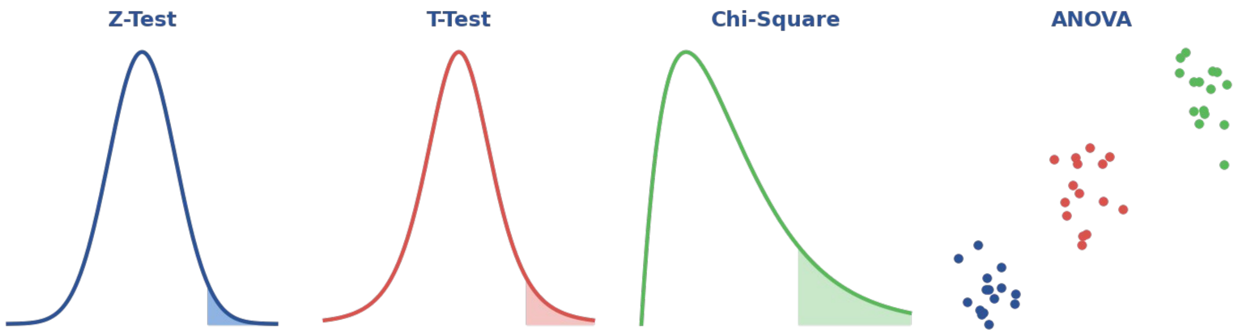
### *3. Chi-square test*

Checks whether two categorical variables are associated

### *4. ANOVA test*

Analysis of variance

Used to compare the mean of three or more groups

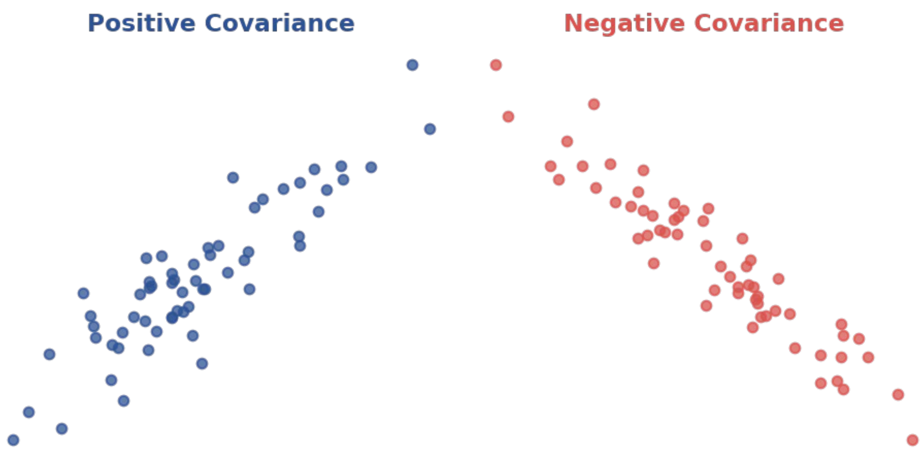



## 7. What is Covariance?

Covariance measure how to variables change together

Positive => Both increase

Negative => One increase while the other decreases




## 8. What is Correlation?

Correlation measure the strength and direction of the relationship between two variable

Rang = -1 to +1

+1 => Perfect Positive

0 => No relationship

-1 => Perfect Negative